PRACTICAL 3
CREATE BUCKETS IN CLOUD STORAGE

Practical objective:
In this practical, you will create, configure, upload to, inspect, and delete a Google Cloud Storage bucket using safe beginner-friendly settings.

Cloud platform used:
Google Cloud Platform, also called GCP.

Main service used:
Cloud Storage.

Estimated outcome:
By the end of this practical, you will be able to:
- Open Cloud Storage in Google Cloud Console.
- Create a storage bucket.
- Understand bucket name, location, storage class, access control, public access prevention, soft delete, and labels.
- Upload an object to a bucket.
- View object details.
- Create a folder-like object prefix.
- Download and delete objects.
- Delete a bucket after practical completion.
- Review bucket configuration using current Google Cloud Storage best practices.

Important safety message:
Cloud Storage can create charges for stored data, operations, retrieval, and network transfer. For lab work, upload only small test files and delete the bucket after the practical unless your teacher instructs you to keep it.

1. BASIC THEORY BEFORE STARTING

1.1 What is Cloud Storage?

Cloud Storage is Google Cloud's object storage service. It is used to store files as objects inside buckets.

Cloud Storage can store:
- Text files
- Images
- Videos
- Backups
- Logs
- Static website files
- Datasets
- Application uploads

Cloud Storage is not the same as a normal folder on your laptop. It stores data as objects inside buckets and is designed for cloud-scale storage.

1.2 What is a bucket?

A bucket is the main container used to store objects in Cloud Storage.

Important points:
- Every object must be inside a bucket.
- Bucket names must be globally unique.

- Buckets cannot be nested inside other buckets.
- A bucket has a location and default storage class.

1.3 What is an object?

An object is a file stored in a Cloud Storage bucket.

An object includes:
- File data
- Object name
- Metadata
- Storage class
- Generation/version information

When you upload a file such as notes.txt or image.png to a bucket, it becomes an object.

1.4 What is a bucket name?

A bucket name is the globally unique name assigned to a Cloud Storage bucket.

Bucket name rules include:
- Must be globally unique across Cloud Storage.
- Usually use lowercase letters, numbers, and hyphens.
- Must start and end with a letter or number.
- Should not contain sensitive information.

Example:
cloudnative-practical-3-yourname-2026

Because bucket names are globally unique, a simple name like test-bucket is probably already taken. Add your project name, initials, or random number.

1.5 What is bucket location?

Bucket location controls where object data is stored.

Common location types:
- Region
- Dual-region
- Multi-region

Region:
- Data is stored in one specific region.
- Example: us-central1.

Dual-region:
- Data is stored in two selected regions.

Multi-region:
- Data is stored across a large geographic area such as US, EU, or ASIA.

For practical work, a single region is usually enough. For real systems, location affects latency, availability, compliance, and cost.

1.6 What is storage class?

Storage class controls cost and access pattern for objects.

Common storage classes:

Standard:
- Best for frequently accessed data.

Nearline:
- Best for data accessed about once a month.

Coldline:
- Best for data accessed about once a quarter.

Archive:
- Best for rarely accessed long-term data.

For this practical, use Standard storage because lab files are temporary and may be accessed immediately.

1.7 What is Uniform Bucket-Level Access?

Uniform Bucket-Level Access, or UBLA, uses IAM permissions for the whole bucket instead of object-level ACLs.

Uniform Bucket-Level Access is recommended because it simplifies permissions and reduces the chance of accidental data exposure.

1.8 What is Public Access Prevention?

Public Access Prevention blocks public access to buckets and objects.

It prevents access through:
- allUsers
- allAuthenticatedUsers

Public access prevention helps stop accidental public exposure. Keep it enabled unless your teacher specifically asks you to create a public website bucket.

1.9 What is soft delete?

Soft delete keeps recently deleted objects or buckets recoverable for a retention period.

1.10 What is lifecycle management?

Lifecycle management automatically performs actions on objects based on rules.

Examples:
- Delete objects older than 30 days.
- Move old objects to Nearline or Coldline storage.
- Delete temporary files with a specific prefix.

Lifecycle rules help reduce cost and clean up old data automatically.

2. REQUIREMENTS

2.1 Hardware requirements

- Computer or laptop
- Internet connection

2.2 Software requirements

- Web browser
- Optional: Cloud Shell for command-line method

2.3 Cloud requirements

- Google Cloud account
- Google Cloud project
- Billing enabled if required
- Permission to create Cloud Storage buckets

Required permission concept:
To create and manage buckets, a user commonly needs a role such as Storage Admin or appropriate custom permissions assigned by the administrator.

If you cannot create a bucket, your account may not have enough IAM permissions or billing might not be enabled for the project.

3. STEP-BY-STEP PRACTICAL

STEP 1: Open Google Cloud Console

Open:

https://console.cloud.google.com

Sign in with your Google Cloud account.

Use the same account assigned for your lab work so that your permissions, project, and billing settings are correct.

Expected result:
Google Cloud Console opens successfully.

STEP 2: Select the correct project

Use the project selector at the top of the page and select the project for this practical.

Buckets are created under a project. Selecting the wrong project can create confusion and billing issues.

Expected result:
The correct project is selected.

STEP 3: Open Cloud Storage

In the search bar, type:

Cloud Storage

Open Cloud Storage.

Cloud Storage is the Google Cloud service used to create buckets and store objects.

Expected result:
The Cloud Storage page opens.

STEP 4: Open Buckets page

In Cloud Storage, open:

Buckets

The Buckets page shows all buckets in the selected project.

Expected result:
The buckets list page opens.

STEP 5: Click Create

Click:

Create

or

Create bucket

This opens the bucket creation form where you will configure name, location, storage class, access control, and protection options.

Expected result:
The bucket creation page opens.

STEP 6: Enter a globally unique bucket name

Enter a bucket name such as:

cloudnative-practical-3-yourname-2026

Replace yourname with your name, initials, roll number, or another unique value.

Bucket names are globally unique across all Google Cloud users. If your chosen name is already taken, change it by adding your initials or a random number.

Important warning:
Do not include private information such as phone number, password, email address, or secret project name in a bucket name.

Expected result:
A valid unique bucket name is entered.

STEP 7: Choose bucket location type

Choose:

Region

Then select a region instructed by your teacher.

Example:

us-central1

For lab work, regional storage is simple and usually sufficient. Real-world systems choose location based on users, compliance, availability, and cost.

Expected result:
Bucket location is selected.

STEP 8: Choose default storage class

Select:

Standard

Standard storage is suitable for frequently accessed data. Since lab files are small and temporary, Standard is a good beginner option.

Expected result:
Standard storage class is selected.

STEP 9: Configure access control

Choose:

Uniform bucket-level access

if the option is shown.

Uniform bucket-level access is recommended because IAM controls access for the bucket and objects consistently.

Expected result:
Uniform bucket-level access is selected or enabled.

STEP 10: Configure public access prevention

Choose:

Enforce public access prevention

or keep public access prevention enabled if it is already inherited.

Public access prevention helps ensure your bucket and objects are not accidentally made public on the internet.

Expected result:
Public access prevention is enforced or inherited from project/organization policy.

STEP 11: Review object protection settings

If soft delete is shown, keep the default soft delete protection or configure it as instructed by your teacher.

Soft delete protects against accidental deletion. It can help recover deleted objects or buckets for a limited time.

Expected result:
Object protection settings are reviewed.

STEP 12: Add labels

Add labels such as:

Key: purpose
Value: practical

Key: owner
Value: student

Labels help organize resources and identify why a resource was created.

Expected result:
Labels are added to the bucket.

STEP 13: Create the bucket

Review all settings and click:

Create

After clicking Create, Google Cloud creates the bucket with your selected configuration.

Expected result:
The bucket is created and appears in the bucket list.

STEP 14: Open the bucket

Click the bucket name in the bucket list.

Opening the bucket lets you upload, view, download, and delete objects.

Expected result:
The bucket details page opens.

STEP 15: Create a small test file on your computer

Create a small text file on your computer named:

cloudnative_test.txt

Add this content:

This is a test file for Cloud Storage Practical 3.

Use a small text file for the practical. Avoid uploading personal, private, or large files.

Expected result:
A small test file is ready for upload.

STEP 16: Upload the test file

Inside the bucket, click:

Upload files

Select cloudnative_test.txt and upload it.

Uploading a file stores it as an object inside the bucket.

Expected result:
The file appears in the object list.

STEP 17: Open object details

Click the uploaded object name.

Observe details such as:
- Object name
- Size
- Storage class
- Created time
- URL or gsutil URI

Object details help you understand how Cloud Storage tracks metadata for each uploaded file.

Expected result:
Object metadata is visible.

STEP 18: Copy the object URI

Find the object URI, usually in this format:

gs://bucket-name/cloudnative_test.txt

The gs:// URI is used by Google Cloud tools to identify Cloud Storage objects.

Expected result:
The object URI is copied or written in your observation table.

STEP 19: Download the object

Use the download option to download the file back to your computer.

Downloading confirms that the object was stored successfully and can be retrieved

when needed.

Expected result:
The test file is downloaded successfully.

STEP 20: Create a folder-like prefix

In the bucket, click:

Create folder

Create a folder named:

practical-files

Cloud Storage does not use folders exactly like a normal file system. Folder names are usually object name prefixes that help organize objects.

Expected result:
A folder-like prefix appears in the bucket.

STEP 21: Upload another test file into the folder

Open the practical-files folder and upload another small text file.

Example file name:

notes.txt

This shows how object prefixes can organize files inside a bucket.

Expected result:
The second file appears under practical-files.

STEP 22: Review permissions

Open the Permissions tab for the bucket.

Do not add public access.

Permissions control who can read, write, or manage bucket data. Avoid granting access to allUsers or allAuthenticatedUsers.

Expected result:
Bucket permissions are reviewed and no public access is added.

STEP 23: Review lifecycle option

Find lifecycle management options if available in the console.

Do not create complex lifecycle rules unless your teacher asks you.

Optional simple rule concept:
Delete temporary objects older than a selected number of days.

Lifecycle rules are useful for automatically cleaning old data and reducing storage cost.

Expected result:
Lifecycle management option is identified.

STEP 24: Delete uploaded objects

Delete:
- cloudnative_test.txt
- notes.txt
- practical-files folder or prefix if shown

A bucket must usually be empty before it can be deleted. Delete test objects first.

Expected result:
The bucket becomes empty.

STEP 25: Delete the bucket

Go back to the bucket list, select your bucket, and click:

Delete

Confirm deletion by typing the required text if asked.

Deleting the bucket prevents unnecessary storage charges after the practical. Remember that bucket names can potentially be reused by others after deletion.

Expected result:
The bucket is deleted.

STEP 26: Confirm cleanup

Refresh the Buckets page and confirm that the practical bucket is no longer listed.

Always verify cleanup. Cloud cost problems often happen because resources are created and forgotten.

Expected result:
No practical bucket remains in the bucket list.

4. OPTIONAL COMMAND-LINE METHOD

This section is optional and should be used only if your teacher wants you to use Cloud Shell.

4.1 Open Cloud Shell

Open Cloud Shell from the Google Cloud Console.

Cloud Shell includes the gcloud storage command, which can manage buckets and objects from the command line.

4.2 Check active project

Run:

```
gcloud config get-value project
```

Always verify the active project before creating cloud resources from the command line.

4.3 Create a bucket

Replace BUCKET_NAME with a globally unique name.

Command:

```
gcloud storage buckets create gs://BUCKET_NAME \
  --location=us-central1 \
  --default-storage-class=STANDARD \
  --uniform-bucket-level-access \
  --soft-delete-duration=7d
```

4.4 Upload a file

Run:

```
echo "Cloud Storage Practical 3 test file" > cloudnative_test.txt
gcloud storage cp cloudnative_test.txt gs://BUCKET_NAME/
```

The cp command copies a local file into the bucket.

4.5 List objects

Run:

```
gcloud storage ls gs://BUCKET_NAME/
```

4.6 Delete object

Run:

```
gcloud storage rm gs://BUCKET_NAME/cloudnative_test.txt
```

Objects must be deleted before deleting the bucket.

4.7 Delete bucket

Run:

```
gcloud storage buckets delete gs://BUCKET_NAME
```

This removes the bucket. Use this carefully and only for your practical bucket.

5. OBSERVATION TABLE

Write your observations in this format:

Project ID:

Bucket name:

Location type:

Region:

Storage class:

Uniform bucket-level access enabled:

Public access prevention enabled or inherited:

Soft delete setting observed:

Labels added:

Uploaded object name:

Uploaded object size:

Object URI:

Was the object downloaded successfully?

Was the bucket deleted after practical?

6. RESULT

A Cloud Storage bucket was created successfully. A test file was uploaded, viewed, downloaded, and deleted. Bucket permissions and configuration options were reviewed, and the bucket was deleted after the practical.

7. CONCLUSION

This practical covered how to create and manage a Google Cloud Storage bucket. They also learned the importance of globally unique bucket names, storage location, storage class, Uniform Bucket-Level Access, public access prevention, labels, lifecycle awareness, and cleanup.

8. LATEST-PRACTICES REVIEW TEST

This review checks whether the practical follows current recommended Google Cloud Storage practices.

1. Was the bucket name globally unique and free from sensitive information?
   Yes. Bucket names are globally visible in the Cloud Storage namespace and should not include private information.

2. Was a suitable location selected?
   Yes. For lab work, a single region is usually enough. Real workloads should choose location based on users, compliance, availability, and cost.

3. Was Standard storage class used for this lab?
   Yes. Standard is suitable for frequently accessed or temporary lab data.

4. Was Uniform Bucket-Level Access enabled?
   Yes. UBLA is recommended because IAM is simpler and safer than managing both IAM and object ACLs.

5. Was public access prevention enabled or inherited?
   Yes. Public access prevention helps avoid accidental exposure of bucket data.

6. Was public access avoided?
   Yes. The practical does not require public access. Do not grant access to allUsers or allAuthenticatedUsers.

7. Was soft delete reviewed?
   Yes. Soft delete helps protect against accidental deletion for a retention period.

8. Were labels added?
   Yes. Labels help identify resource purpose, owner, and cost.

9. Was lifecycle management identified?
   Yes. Lifecycle rules help automatically delete or transition old objects to optimize cost.

10. Were test objects and the bucket deleted after the practical?
   Yes. Cleanup avoids unnecessary storage charges.

If the points above are covered, the practical aligns with current platform practices.
practices and introduces current bucket security and cost-management recommendations.

9. VIVA QUESTIONS

1. What is Cloud Storage?
2. What is a bucket?
3. What is an object?
4. Why must bucket names be globally unique?
5. What is a storage class?
6. What is Standard storage class used for?
7. What is Uniform Bucket-Level Access?
8. What is Public Access Prevention?
9. What is soft delete?
10. What is lifecycle management?
11. Why should public access be avoided for normal lab buckets?
12. Why should test buckets be deleted after practical?

10. SHORT ANSWERS FOR VIVA

1. Cloud Storage is Google Cloud's object storage service.
2. A bucket is a container that stores objects.
3. An object is a file stored in a bucket.
4. Bucket names are globally unique because all Cloud Storage users share one bucket namespace.
5. A storage class defines cost and access pattern for stored objects.
6. Standard storage is used for frequently accessed data.
7. Uniform Bucket-Level Access manages access using IAM at the bucket level.
8. Public Access Prevention blocks public access through allUsers and allAuthenticatedUsers.
9. Soft delete keeps deleted data recoverable for a limited retention period.
10. Lifecycle management automatically performs actions such as deleting old objects or changing storage class.
11. Public access should be avoided to prevent accidental data exposure.
12. Test buckets should be deleted to avoid unnecessary cloud charges.

11. COMMON MISTAKES TO AVOID

1. Using a bucket name that is already taken.
2. Including private information in the bucket name.
3. Creating the bucket in the wrong project.
4. Selecting an expensive or unsuitable location without understanding cost.
5. Disabling Uniform Bucket-Level Access without reason.
6. Allowing public access accidentally.
7. Uploading personal or sensitive files for a lab test.
8. Forgetting to delete uploaded objects.
9. Forgetting to delete the bucket.
10. Ignoring lifecycle and cost-management options.

12. PRACTICAL COMPLETION CHECKLIST

Before submitting this practical, confirm:

1. You opened Google Cloud Console.
2. You selected the correct project.
3. You opened Cloud Storage.
4. You opened the Buckets page.
5. You created a bucket with a globally unique name.
6. You selected a location.
7. You selected Standard storage class.
8. You enabled or confirmed Uniform Bucket-Level Access.
9. You enabled or confirmed Public Access Prevention.
10. You reviewed soft delete.
11. You added labels.
12. You uploaded a small test file.
13. You viewed object details.
14. You copied or noted the object URI.
15. You downloaded the object.
16. You created a folder-like prefix.
17. You reviewed permissions.
18. You identified lifecycle management.
19. You deleted test objects.
20. You deleted the bucket.
21. You completed the latest-practices review test.

22. You answered the viva questions.